

DINOv2-reg ViT-B/14

256 × 768 (16 x 16 patches)

Drop token 25%

Layer Normalization

MultiHead Attention 4 heads

Layer Normalization

Linear 256 → 1024 ↓ GELU ↓ Linear 1024 →256

K, V from the other stream

Clinical metadata:

age, gender, region, pain, itch, diameter, ...

CLIP ViT-B/32 tex

77 × 512 (token features)

Drop text 30%

Layer Normalization

MultiHead Attention 4 heads

Layer Normalization

Linear 256 → 1024 ↓ GELU ↓ Linear 1024 →256

Transformer encoder layer ×2

Layer Normalization

K Q V

Scaled dot-product attention, 4 heads

Concat heads→ Dropout(0.15)

Layer Normalization

Linear 256 → 1024 ↓ GELU → Dropout(0.15) ↓ Linear 1024 → 256 → Dropout(0.15)

Final Layer Normalization

select CLS row

Linear 256 → 256 ↓ LayerNorm → GELU ↓ Linear 256→ 256

Linear 256 → 128 ↓ LayerNorm → GELU ↓ Linear 128 → 64

Wrapped-Gaussian likelihood, logits & hierarchical loss

Metadata prior $\mathbf{p}^2_c$

MLP: LN 256 → GELU → 64

clamp [-14, 3]

Log Determinant term

Bayesian shrinkage (conjugate variance posterior) Data spread shrunk towards the prior

Hybrid Support Spread

Own class coordinates

Metadata conditioned Gaussian Prototype head

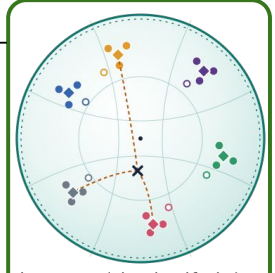
Mahalanobis term

Wrapped Normal log-likelihood

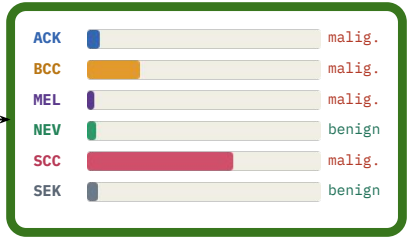
Tangent Coordinates Möbius addition every class c, every point x

Einstein midpoint of each class's supports (computed in the Klein model)

Back to the ball ← Weighted Mean ← Lorentz Factor ← Klein Map



◆ prototype (Einstein midpoint)
• support ○ query × one query
Dashed curves: geodesics, bowing toward 0



Tanh gate
Element-wise Addition
Frozen
Image Stream
Text Stream